

Symplectic Neural Flows for Modelling and Discovery

FoCM 2026

Workshop on Geometric Integration and Computational Mechanics

Davide Murari

Department of Applied Mathematics and Theoretical Physics, University of Cambridge

Joint work with Priscilla Canizares, Carola-Bibiane Schönlieb, Ferdia Sherry, and Zakhar Shumaylov

Vienna, 15 July 2026

Flow maps in modern deep learning

Generative modelling

Learn transport from a simple distribution to data:

- ▶ normalising flows,
 - ▶ flow matching,
 - ▶ flow map matching.
- ▷ Lipman et al. 2022, *Flow Matching for Generative Modeling*
- ▷ Boffi et al. 2024, *Flow map matching with stochastic interpolants: A mathematical framework for consistency models*

Theory of deep learning

Expressivity through compositions of flow maps:

$$x \mapsto \phi_T(x).$$

This gives a language for approximation, controllability, and depth.

- ▷ Li et al. 2019, *Deep Learning via Dynamical Systems: An Approximation Perspective*
- ▷ Ruiz-Balet and Zuazua 2023, *Neural ODE control for classification, approximation and transport*

Continuous-depth models

Replace a finite stack of layers by a trainable ODE:

$$\dot{z} = f_\theta(t, z),$$

$$\phi_T^\theta(z_0) = z(T).$$

Structure can be encoded in f_θ .

- ▷ Chen et al. 2018, *Neural Ordinary Differential Equations*
- ▷ Celledoni et al. 2020, *Structure preserving deep learning*

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This talk focuses on symplectic neural flow maps:
continuous-depth models with approximation-theoretic support.

Machine learning for Hamiltonian dynamics

Learn the mechanics

- ▶ Hamiltonian Networks.
- ▶ Lagrangian Networks and forced systems.
- ▷ Greydanus et al. 2019, *Hamiltonian Neural Networks*
- ▷ Dierkes et al. 2023, *Hamiltonian Neural Networks with Automatic Symmetry Detection*
- ▷ Hansen et al. 2025, *Learning mechanical systems from real-world data using discrete forced Lagrangian dynamics*

Learn symplectic maps

- ▶ SympNets.
- ▶ HénonNet.
- ▶ Learned maps as numerical surrogates.
- ▷ Jin et al. 2020, *SympNets: Intrinsic structure-preserving symplectic networks for identifying Hamiltonian systems*
- ▷ Drimalas et al. 2025, *Symplectic neural network and its application to charged particle dynamics in electromagnetic fields*

Solve Hamiltonian equations with NNs

- ▶ PINN-style residual losses.
- ▶ From known equations to approximate solutions.
- ▷ Mattheakis et al. 2020, *Hamiltonian neural networks for solving equations of motion*

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We combine the three perspectives by designing a neural network that is the exact flow of a time-dependent Hamiltonian, and train it from either equations or trajectory data.

Hamiltonian flow maps

Let $H : \mathbb{R}^{2d} \rightarrow \mathbb{R}$ be twice continuously differentiable and

$$\dot{x} = \mathbf{J} \nabla H(x), \quad x = (q, p), \quad \mathbf{J} = \begin{pmatrix} 0 & \mathbf{I}_d \\ -\mathbf{I}_d & 0 \end{pmatrix}.$$

Flow map

$$\begin{aligned} \frac{d}{dt} \phi_{H,t}(x_0) &= \mathbf{J} \nabla H(\phi_{H,t}(x_0)), \\ \phi_{H,0}(x_0) &= x_0. \end{aligned}$$

Symplecticity

$$D\phi_{H,t}(x)^\top \mathbf{J} D\phi_{H,t}(x) = \mathbf{J}.$$

Energy conservation

$$H(\phi_{H,t}(x_0)) = H(x_0), \quad t \geq 0.$$

First learning task: Fit trajectory data

What we observe

Trajectory samples from an unknown Hamiltonian system:

$$y_m^n = \phi_{H,t_m^n}(x_0^n) + \varepsilon_m^n,$$

$t_m^n \in [0, \Delta t]$ (non-uniform) and $x_0^n \in \Omega$.

What we learn

A continuous-time map

$$\bar{\psi} : [0, \Delta t] \times \Omega \rightarrow \Omega, \quad \bar{\psi}_t(x_0) \approx \phi_{H,t}(x_0),$$

then extend it in time by composition.

Supervised objective

$$\mathcal{L}_{\text{sup}}(\bar{\psi}) = \frac{1}{NM} \sum_{n=1}^N \sum_{m=1}^M \|\bar{\psi}_{t_m^n}(x_0^n) - y_m^n\|_2^2.$$

Residual objective

$$\mathcal{L}_{\text{res}}(\bar{\psi}) = \frac{1}{N} \sum_{n=1}^N \left\| \partial_t \bar{\psi}_{t_n}(\mathbf{x}_0^n) - \mathbf{J} \nabla H(\bar{\psi}_{t_n}(\mathbf{x}_0^n)) \right\|_2^2.$$

What we sample

Collocation points

$$(t_n, \mathbf{x}_0^n) \in [0, \Delta t] \times \Omega.$$

The target vector field is known through H .

Short-time learning, long-time use

- Assume a compact forward invariant set $\Omega \subset \mathbb{R}^{2d}$.

- Learn a map on a short window:

$$\bar{\psi} : [0, \Delta t] \times \Omega \rightarrow \Omega, \quad \bar{\psi}_t \approx \phi_{H,t}.$$

- Roll out by composition. With $k = \text{floor}(t/\Delta t)$ and $r = t - k\Delta t$,

$$\psi(t, x_0) = \bar{\psi}_r \circ (\bar{\psi}_{\Delta t})^k (x_0).$$

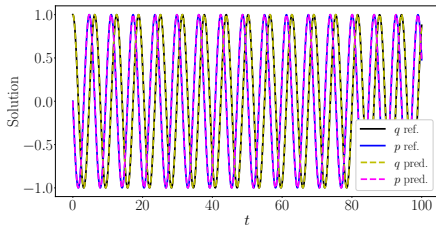


Figure: Short-window network applied repeatedly to $T = 100$.

A geometric-integration perspective

- ▶ Geometric integration: if the flow has a qualitative property, the numerical method should preserve it.
- ▶ This structure-conservation typically improves long-time behaviour.
- ▶ A generic neural network can fit short trajectories while changing the qualitative dynamics.
- ▶ A learned long-time simulator should respect the same geometry as the underlying flow map.

$$H(\phi_{H,t}(x_0)) = H(x_0)$$
$$D\phi_{H,t}(x)^\top \mathbf{J} D\phi_{H,t}(x) = \mathbf{J}$$

Each layer composes two exact flows of time-dependent Hamiltonians:

$$\begin{aligned}\phi_{q,t}(q,p) &= \begin{pmatrix} p - (\nabla_q V_q(t,q) - \nabla_q V_q(0,q)) \\ q \end{pmatrix} \\ &= \begin{pmatrix} p - \int_0^t \nabla_q \partial_s V_q(s,q) ds \\ q \end{pmatrix}.\end{aligned}$$

$$\phi_{p,t}(q,p) = \begin{pmatrix} q + (\nabla_p V_p(t,p) - \nabla_p V_p(0,p)) \\ p \end{pmatrix}.$$

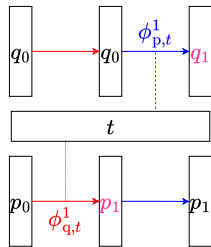


Figure: One SympFlow layer.

With L layers,

$$\bar{\psi}_t = \phi_{p,t}^L \circ \phi_{q,t}^L \circ \cdots \circ \phi_{p,t}^1 \circ \phi_{q,t}^1.$$

Parametrisation

The scalar potentials V_q^i, V_p^i are represented by MLPs depending on (t, q) or (t, p) .

Immediate consequence

$\bar{\psi}_0(x) = x$, and every $\bar{\psi}_t$ is symplectic because it is a composition of exact Hamiltonian flows.

The Hamiltonian behind SympFlow

Composition rule (de Gosson, 2006, Prop. 2.54)

For $\psi_t = \phi_{H^2,t} \circ \phi_{H^1,t}$, the composed map is the exact time- t flow generated by

$$H^3(t, x) = H^2(t, x) + H^1\left(t, \phi_{H^2,t}^{-1}(x)\right).$$

One SympFlow layer

$$\begin{aligned} H_t^i(q, p) &= \partial_t V_p^i(t, p) \\ &\quad + \partial_t V_q^i(t, \tilde{q}), \\ \tilde{q} &= q - \nabla_p V_p^i(t, p) + \nabla_p V_p^i(0, p). \end{aligned}$$

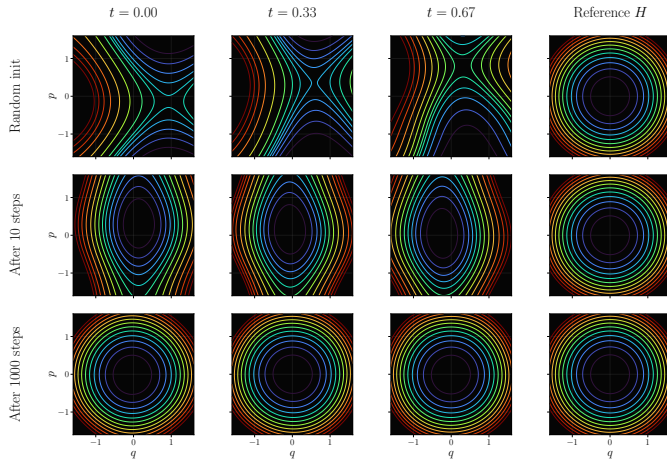
Whole network

$$\begin{aligned} \mathcal{H}(\bar{\psi}) &= H_t^{L:1}, \\ H_t^{L:i} &= H_t^{L:(i+1)} + H_t^i \circ \phi_{H_t^{L:(i+1)},t}^{-1}. \end{aligned}$$

The trained map $\bar{\psi}_t$ and its generating Hamiltonian $\mathcal{H}(\bar{\psi})$ are both accessible.

Hamiltonian of a trained SympFlow

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Universality: the constraint is expressive

Universal approximation theorem for SympFlow

Let $H : \mathbb{R} \times \mathbb{R}^{2d} \rightarrow \mathbb{R}$ be twice continuously differentiable. Let $\Omega \subset \mathbb{R}^{2d}$ be compact and forward invariant for the exact flow $\phi_{H,t}$. For every $\varepsilon > 0$, there is a SympFlow $\bar{\psi}_t$ such that

$$\sup_{\substack{t \in [0, \Delta t] \\ x \in \Omega}} \|\bar{\psi}_t(x) - \phi_{H,t}(x)\|_{\infty} < \varepsilon.$$

The initial time of the two-time flow of $H = H_t$ is set to $t_0 = 0$.

Symplecticity is not obtained by reducing to a small class of maps.

Proof idea - Part I: reduce to separable short-time flows

$$\phi_{H,t} \xrightarrow[\text{Gronwall}]{C^1 \text{ polynomial approximation}} \phi_{\tilde{H},t} \xrightarrow{\text{short-time composition}} F_n \xrightarrow[\text{Turaev + neural approximation}]{\text{separable Hamiltonian flow}} \phi_{\bar{H}_n,1}.$$

Polynomial surrogate

Approximate H and ∇H by $\tilde{H}$ polynomial in x and continuous in t .

Gronwall transfers this to a flow approximation.

Short time steps

Set $h = t/N$, and let F_n be the two-time flow from nh to $(n+1)h$.

$$\phi_{\tilde{H},t} = F_{N-1} \circ \cdots \circ F_0.$$

Turaev extension

Approximate each short map F_n by a separable neural Hamiltonian flow:

$$\bar{H}_n(s, q, p) = \bar{V}_{p,n}(s, p) + \bar{V}_{q,n}(s, q),$$

$$\sup_{x \in \Omega} \|\phi_{\bar{H}_n,1}(x) - F_n(x)\|_{\infty} = \mathcal{O}(h^2).$$

Turaev (2002) + neural approximation.

How do we go from $\phi_{\bar{H}_n,1}$ to the SympFlow architecture?

Proof idea - Part II: Lie–Trotter produces SympFlow layers

$$\text{Fix } n: \bar{H}_n(s, q, p) = T_n(s, p) + V_n(s, q). \text{ On } I_k = [k/K, (k+1)/K], \\ \phi_{\bar{H}_n, I_k} = \Psi_k + \mathcal{O}(K^{-2}), \quad \Psi_k := \phi_{p,k} \circ \phi_{q,k}, \quad \phi_{\bar{H}_n, 1} = \Psi_{K-1} \circ \cdots \circ \Psi_0 + \mathcal{O}(K^{-1}).$$

Combining the approximations

Combining the three approximations gives:

$$\phi_{\tilde{H}, t} = \text{SympFlow} + \mathcal{O}(\Delta t^2/N) + \mathcal{O}(N/K), \quad K = N^2.$$

Splitting gives the error; layer representation is exact.
The polynomial flow can be chosen arbitrarily close to the original flow.

Equations of motion

$$\dot{x} = p_x, \quad \dot{y} = p_y, \quad \dot{p}_x = -x - 2xy, \quad \dot{p}_y = -y - (x^2 - y^2).$$

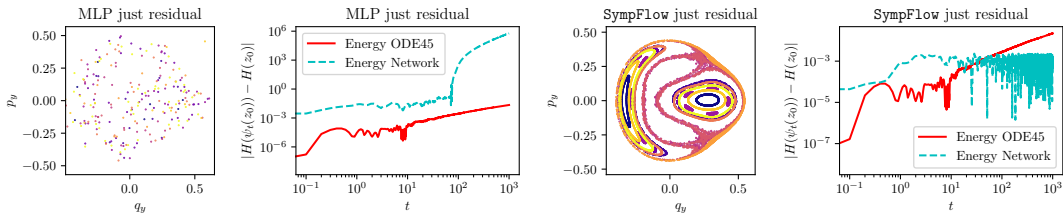


Figure: **Unsupervised experiment — Hénon–Heiles:** Comparison of the Poincaré sections and the energy behaviour up to time $T = 1000$.

Damped harmonic oscillator (supervised)

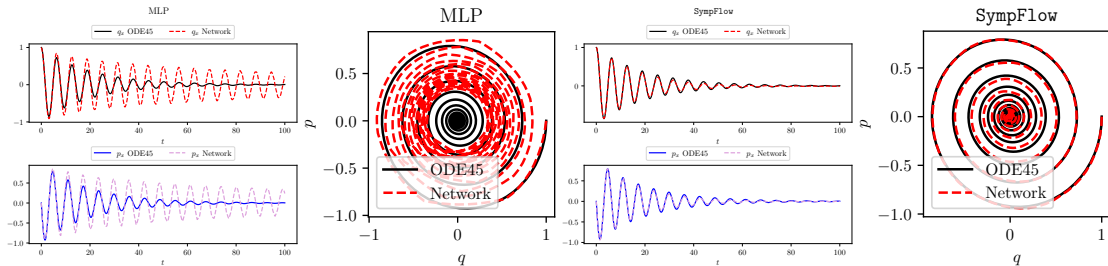


Figure: **Supervised experiment — damped harmonic oscillator:** $N = 50$, $M = 10$, no noise ($\varepsilon = 0$), damping $\lambda = 0.1$.

Reference for non-conservative dynamics:

Galley (2013), *Classical mechanics of nonconservative systems*, Physical Review Letters.

Is SympFlow any better than symplectic integrators?

Not right now.

If the goal is simulating high-dimensional (or) expensive multi-query Hamiltonian systems, SympFlow is a promising surrogate.

We are exploring this question with Priscilla Canizares in the context of numerical relativity.

Is there more theory to be done?

Sure!

How can we model the SympFlow layers to ensure they are at least a consistent solver of the underlying ODE when it is known? What can we say about the long-time energy behaviour? ...

Thank you for the attention

Questions?

Canizares, P., Murari, D., Schönlieb, C. B., Sherry, F., and Shumaylov, Z.
Symplectic Neural Flows for Modelling and Discovery.
SIAM Journal on Scientific Computing, in press.

davidemurari.com/sympflow